

Chen-Yu Chang

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EDUCATION

Carnegie Mellon University | Pittsburgh, PA

Dec 2027

Master of Science in Civil and Computer Engineering

Boston University | Boston, MA

May 2022

Bachelor of Science in Computer Engineering, Minor: Statistics

EXPERIENCE

Product Marketing Engineer | *Texas Instruments Sugar Land, TX*

Sept 2022 - Jun 2023, Jan 2024 - Feb 2025

- Expanded market share into new automotive segments (lighting, braking, BMS, zonal architecture), driving a 54% revenue increase through direct sales engagement with worldwide OEMs
- Launched three products (F280015x, F28P55x, AM261x) end-to-end, crafting go-to-market campaigns around application use cases (including OBC, DCDC, traction) and competitive differentiation, reaching more than 50M revenue with 80+ design wins
- Proposed an Ethernet-based zonal architecture presentation with Tesla (Ether-Ring AM26x) that cut wiring weight by 68% and assembly cost by 13.5%; demoed the solution at *CES 2025*
- Visited 7 automotive OEMs and Tier1s in person to define next-generation MCU requirements, securing oral commitments from accounts representing 50M+ units per year and shape future EV architecture
- Directed definition of unreleased MCU platforms by synthesizing customer feedback on peripherals and performance requirements for next-generation vehicle models, including F29 platform and new integration platforms
- Hosted a TI/MathWorks webinar on Model-Based Design to 357 attendees, presenting TIDM-02012 HVAC reference design
- Delivered technical presentations at *Electronica Europe* on C29 CPU performance benchmarks and their impact on system cost and development ecosystems

Applications Engineer | *Texas Instruments Sugar Land, TX*

Jun 2023 - Dec 2023

- Spearheaded launch of the F29x product line, authoring the datasheet and whitepaper; whitepaper nominated to present at *Electronica Europe* on C29 CPU performance benchmarks and their impact on system cost and development ecosystems
- Collaborated with senior engineers to benchmark C29x vs C28x and competitor cores, demonstrating 4.31x faster signal-chain execution, 3–4x improvement in real-time control, and near-zero-cycle interrupt discontinuity overhead through delayed-branch architecture
- Provided technical support to customers on C2000 peripherals (DCSM, PWM) via TI's corporate E2E platform
- Designed a testing protocol temperature testing procedures for F28P65x die, validating 23 units per day and enabling the product's release by confirming defect-free silicon

PROJECTS

Beyond Air Tapping | *Client: Rtangent (Flutter, Unity, Python, Swift, AR) — BU Capstone*

Sep 2021 - May 2022

- Built an AR system that projected 3D models from a QR-coded hardware baton, rendered live through an iOS viewer app
- Developed an iOS admin app with authenticated access control for uploading and managing project content per QR code
- Programmed dynamic QR code switching via hardware button input, enabling on-demand 3D model rotation and multi-angle viewing

Man in the Middle Attack: Principles and Tricks — *Course Research*

Oct 2021 – Dec 2021

- Implemented DoS, ARP spoofing, DNS spoofing, and SSL stripping attacks using Kali Linux, Ettercap, and Wireshark
- Simulated live credential-capture attacks on unsecured networks to evaluate HTTP vs. HTTPS vulnerability gaps
- Proposed defense strategies including VPN deployment, dynamic ARP inspection, HSTS preload, and static ARP tables

Graph Traversal: Breadth-First Search (BFS) Optimization (C, CUDA) — *Course Research*

Feb 2021 - May 2021

- Benchmarked serial, hybrid, multithreaded, and GPU-accelerated BFS implementations, achieving up to 144x speedup over baseline
- Improved performance via top-down/bottom-up switching heuristics, cache blocking, and CUDA-based parallelization

Cruise Control (C, Node.js) — *Course Project*

Sep 2020 - Dec 2020

- Integrated ultrasonic, LiDAR, and optical encoder sensors with steering servos on an ESP32 for closed-loop autonomous navigation
- Tuned a PID controller over UDP and Socket.io to maintain stable speeds of 0.1–0.4 m/s with real-time obstacle avoidance
- Pushed the platform to 1 m/s under stress testing while maintaining smooth, failure-free motion control

SKILLS

Languages: Advanced - C++, Python; Intermediate - C, CUDA, JavaScript, MATLAB, Swift, Verilog

Tools & Frameworks: Node.js, Flutter, React Native, Unity, Eagle CAD, VS Code, Xcode, Confluence

Hardware & Systems: Sensor integration, Embedded systems, Machine learning, AR/VR development, Computer Vision

LEADERSHIP

Cofounder, Broccoli's in the Fridge

Sept 2020 - Jan 2022

- Cofounded a food-waste mitigation startup, funded by BU's Sustainability Seed Grant and recognized university-wide for social impact
- Built an iOS app in Swift with receipt-image upload, best-use-date tracking, and recipe suggestions to reduce household food waste

Hyper Rocket Team

Oct 2020 - Jan 2021

- Collaborated on the design of a high-power rocket targeting a 3,000m apogee and optimized component weight, structural design, and electrical systems through simulation to maximize altitude performance
- Led the electrical team in designing a custom PCB in Eagle CAD for selected microcontrollers and sensors